

Integrating Graph Algorithms and Dynamic Programming for Inventory, Routing and Resource Decisions in Smart Warehouses: A Literature-Grounded Framework for a Smart Warehouse Inventory Management Platform

*Conceptual research paper (framework proposal). No experimental results of the proposed framework are reported. Evidence labels used throughout: **[P]** = result reported in published/pre-published research; **[F]** = proposed future experiment or analytical argument by the present study.*

1. Abstract

Order picking is widely described as the most labour-intensive and costly warehouse activity; de Koster et al. report estimates of up to 55% of total warehouse operating expense [1]. Exact dynamic programming (DP) for picker routing in rectangular warehouses has been known since 1983 [2] and is still being refined [12]–[14], while storage assignment, replenishment and inventory allocation are generally treated as separate problems [1], [5]–[7]. This study reviews 18 warehouse-related sources whose bibliographic records were verified, identifies a **potential** research gap, and proposes a *Smart Warehouse Inventory Management Platform* framework. The framework (i) represents the warehouse as a weighted graph $G = (V, E)$, (ii) derives location-dependent routing costs with Dijkstra's algorithm, (iii) uses 0/1 knapsack DP to select inventory for capacity-limited forward storage, and (iv) uses a multiple-choice resource-allocation DP to distribute scarce resources across zones. Mathematical formulations, recurrences and complexity bounds are given. Breadth-first and depth-first search and minimum spanning tree algorithms (Prim, Kruskal) are included only where a logically justified role exists, and that justification is labelled as the author's argument rather than published evidence. An evaluation protocol based on synthetic warehouse graphs and a published open instance set [23] is specified; no results are claimed.

2. Keywords

Smart warehouse; inventory management; order picking; warehouse graph model; Dijkstra's algorithm; dynamic programming; 0/1 knapsack; resource allocation; storage assignment; Industry 4.0.

3. Introduction

Warehouses have shifted from passive storage nodes to technology-enriched fulfilment systems. Boysen and de Koster trace this evolution through three generations, ending with robotised distribution centres, and outline a research agenda for the next decades [11]. E-commerce intensifies the pressure: orders are small, numerous and time-critical, which motivates mixed-shelves storage, dynamic order processing and automation [3]. Reviews of smart warehouses catalogue enabling technologies (IoT, robotics, digital twins) and transition challenges [9], [10], [18], but they are descriptive about *which algorithms* should convert the resulting data into decisions.

Two observations motivate this paper. First, a rich body of operations-research work gives exact or near-exact algorithms for individual decisions, notably picker routing by DP [2], [12]–[14]. Second, authoritative reviews state that combinations of warehouse planning problems have hardly been explored [1] and that optimising planning problems sequentially may yield suboptimal overall performance [5]. A lightweight algorithmic layer that connects *spatial* information (the layout graph) with *inventory* decisions (what to stock where, and with which resources) is therefore a sensible object of study, particularly for small and medium warehouses in which commercial optimisation solvers are impractical.

Research question (refined after the literature review). *How can shortest-path computation on a warehouse graph be combined with capacity-constrained dynamic programming so that storage-space selection and resource allocation decisions account for travel cost, and what are the computational limits of such a combination?*

Contributions (stated cautiously). This study proposes (a) a unified formulation that couples graph-derived travel costs with knapsack-type inventory selection and multiple-choice resource allocation; (b) a decomposition heuristic with explicit complexity bounds; (c) a reproducible evaluation protocol that distinguishes published evidence from proposed experiments. It does not claim a first-of-its-kind system, and it does not report experiments of the proposed framework.

Scope note on sources. Searches were targeted rather than a PRISMA-style systematic review, and were run through open web indexes. Items from IEEE Xplore or the ACM Digital Library could not be verified in this session and are therefore not cited. The "Warehouse Operations Data Structure (WODS)" topic suggested in the brief could not be located under that name in any verifiable publication and is not cited.

4. Background

Warehouse model. In picker-to-parts systems, workers travel through aisles and cross-aisles to collect items; order picking is then a routing problem on the aisle network. In a rectangular warehouse with a depot, the shortest picking tour is a special case of the NP-hard travelling salesman problem (TSP) in which items are the cities [12]. Ratliff and Rosenthal showed that when cross-aisles exist only at the ends of the aisles, the minimum-time tour can be found by DP with effort linear in the number of aisles [2] [P]. The same DP generalises to any number of cross-aisles but becomes exponential in the number of cross-aisles [12] [P].

Graph representation. The aisle network is naturally a graph: vertices are pick locations, aisle intersections, the depot and docks; edges are traversable aisle segments weighted by length or travel time. Recent DP refinements explicitly use graph representations of rectangular warehouses and edge-configuration states [14] [P].

Why DP and graph algorithms. Shortest-path algorithms (Dijkstra [20]) give exact travel distances on non-negative weighted graphs; DP gives exact solutions to capacity-constrained selection (knapsack) in pseudo-polynomial time [19]. Both are classical, deterministic, auditable and cheap, which is the property sought for a lightweight decision layer.

Smart warehouse. Following the systematic review of van Geest et al., a smart warehouse aims to raise service quality, productivity and efficiency while lowering cost and failures, using technologies such as sensors, automation and analytics [9]. Industry 4.0 technologies are reviewed with respect to sustainability across receiving, storage, picking, packing and shipping by Perotti et al. [10].

5. Related Work / Literature Review

5.1 Surveys and foundations

De Koster et al. review layout design, storage assignment, routing, batching and zoning for manual order picking, and conclude that combinations of these areas had hardly been explored [1]. Van Gils et al. review literature that combines planning problems and conclude that combining them yields substantial efficiency benefits, while sequential optimisation can be suboptimal [5]. Boysen et al. survey warehousing in the e-commerce era, covering automation, mixed-shelves storage and dynamic order processing [3]; Azadeh et al. review robotised systems and argue that storage slotting, batching and routing logic must be revisited for them [4]; Boysen and de Koster give a fifty-year OR perspective [11].

5.2 Exact routing by dynamic programming

Ratliff and Rosenthal's linear-in-aisles DP [2] is the anchor result. Pansart et al. strengthen a sparse mixed-integer formulation and generalise the DP to more cross-aisles, benchmarking against the TSP solver Concorde [12]. Dunn et al. (2026, preprint) modify single-block and two-block DPs to reduce computation without losing optimality [14]. Lorenz et al. (working paper) design an exact DP for the order batching, sequencing and routing problem with release times; in their experiments (two-block warehouse, 3 cross-aisles, 10 aisles, 20 instances of 15 orders per setting) the DP solved 103 of 120 instances to proven optimality, whereas a MIP solver with a one-hour limit solved none to proven optimality [13] [P]. Bianchessi et al. give a DP, two branch-and-bound algorithms and a MILP for a pick-up and delivery order problem in an automated storage/retrieval system, whose complexity status is open (conjectured NP-hard) [15].

5.3 Storage assignment and replenishment

Dijkstra and Roodbergen derive exact expected route-length formulas for four routing methods under any storage location assignment and use DP to determine assignments; their experiments reveal assignment patterns not previously described [6] [P]. Çelik et al. define the storage replenishment routing problem (which forward-slot items to replenish and the worker's route), prove NP-hardness in general, and test an a-priori-routing heuristic on random instances [7] [P]; the instance set is openly archived [23].

5.4 Capacity-constrained allocation (knapsack family)

Pferschy shows that the 0/1 knapsack problem can be solved in $O(nc)$ time and $O(n + c)$ space, where c is the capacity [19] [P]. Sökel formulates retail store replenishment allocation as an extended multidimensional knapsack problem, notes that GAP and MKP are strongly NP-hard, and proposes an $O(N \log N)$ heuristic; on 212,278 order records from one retailer, weighted same-day coverage rose from 24.3% to 37.8% after go-live [16] [P, preprint, before-after design against the operational policy rather than against alternative algorithms].

5.5 Smart-warehouse technologies and automation

Lee et al. propose an IoT-based warehouse management system with computational-intelligence analytics; using data from a case company they report improved productivity, picking accuracy and efficiency, and robustness to order variability [8] [P]. Jiang et al. transform a distribution-warehouse layout into a raster map, precompute and store shortest paths with A* and circular rules, and then assign tasks to two AGV types with a genetic algorithm; they report that the method meets 24-hour operating requirements

[17] [P]. Alam et al. review AI-enhanced digital twins for warehouse logistics, highlighting path planning, task allocation, inventory management and storage assignment, and note the lack of analysis of economic impact [18] [P].

5.6 Synthesis table

Ref.	Problem	Method	Environment / data	Main finding	Limitation	Relevance here
[2] Ratliff & Rosenthal 1983	Picker routing, rectangular warehouse	DP	Two cross-aisles; 50-aisle case solved in about 1 min on a microcomputer	Linear in number of aisles	Restricted layout; item locations given	Foundational routing cost
[12] Pansart et al. 2018	Exact picking tours	Sparse MIP + generalised DP	Compared with Concorde	Two exact algorithms	DP exponential in cross-aisles; arXiv version	Exact routing benchmark
[13] Lorenz et al. 2024	Batching, sequencing, routing with release times	Exact DP + dominance rules	120 synthetic instances, 15 orders each	DP optimal on 103/120; MIP none in 1 h	Single picker; locations predetermined; working paper	Evidence that DP beats generic solvers on structured problems
[14] Dunn et al. 2026	Order picking, 1- and 2-block	Modified DP	Rectangular graphs	Efficiency gains, optimality kept	Preprint; layout-restricted	Graph-state DP
[6] Dijkstra & Roodbergen 2017	Storage location assignment	Route-length formulas + DP	Picker-to-parts, four routing methods	New assignment patterns	Routing methods fixed; no capacity-selection layer	Closest precedent for coupling storage and routing
[7] Çelik et al. 2022	Replenishment routing	A-priori routing heuristic	Random instances, 25–450 pick items	NP-hard; heuristic evaluated	No exact method for large cases	Inventory-routing link; open instances [23]
[16] Sökel 2026	Capacity-constrained allocation	MKP-based heuristic	212,278 orders, 3 warehouses, 772 stores	Coverage 24.3% → 37.8%	Preprint; no head-to-head comparison; network level, not intra-warehouse	Knapsack-type allocation in practice
[19] Pferschy 1999	0/1 knapsack	DP	Theory	$O(nc)$ time, $O(n+c)$ space	Pseudo-polynomial	Complexity basis
[8] Lee et al. 2018	IoT-enabled WMS	Analytics / computational intelligence	Case company	Productivity and accuracy gains	Single case; abstract does not indicate graph-based routing or capacity-constrained selection	Data layer of framework

Ref.	Problem	Method	Environment / data	Main finding	Limitation	Relevance here
[17] Jiang et al. 2023	AGV paths and task assignment	A* + genetic algorithm	Distribution-warehouse grid map	Meets 24 h operation	Metaheuristic stage; no inventory dimension	Two-stage graph-then-optimize pattern
[15] Bianchessi et al. 2026	AS/RS pick-up and delivery orders	DP, B&B, MILP	Computational study	Exact methods compared	Complexity open	Automated systems
[9], [10], [18]	Smart/Industry 4.0 warehouse reviews	Systematic or narrative review	Literature	Technology and challenge maps	Descriptive on algorithms	Context and motivation

6. Research Gap

The following gap statements are supported by the sources cited with each; they are **provisional** because the search was targeted, not systematic.

- **G1 — Routing treats item locations as given.** The exact routing literature [2], [12]–[14] optimises the tour for fixed storage locations; Lorenz et al. explicitly assume predetermined picking positions and themselves point to upstream decisions such as inventory levels and item repositioning as higher-impact areas [13].
- **G2 — Integration is called for but rarely completed.** Reviews state that combinations of planning problems are hardly explored [1] and that sequential optimisation may be suboptimal [5]. Dijkstra and Roodbergen [6] and Çelik et al. [7] couple storage or replenishment with routing, but neither combines a graph-derived location cost with an explicit capacity-constrained *inventory selection* layer and a zone-level *resource-allocation* layer.
- **G3 — Knapsack-type allocation is studied at network level.** The knapsack-based allocation study in the reviewed set operates between a distribution centre and stores [16], not on the intra-warehouse routing graph.
- **G4 — Smart-warehouse reviews are technology-centred.** Reviews [9], [10], [18] map technologies and challenges; Alam et al. identify missing economic analysis [18]. None of the reviewed items provides a transparent, reproducible algorithmic decision layer that consumes IoT-type data.

Potential research gap. *Within the sources verified here, no study was found that combines (i) a weighted warehouse-graph model with Dijkstra-based travel costs, (ii) 0/1 knapsack DP for forward-storage selection, and (iii) multiple-choice resource-allocation DP across zones in one lightweight, reproducible framework with open synthetic benchmarks.* This statement should be re-tested with a PRISMA-style search of Scopus, Web of Science, IEEE Xplore and ACM DL before submission.

7. Problem Formulation

Sets and parameters. Let $G = (V, E, w)$ be a connected, weighted, undirected graph with $V = V_S \cup V_J \cup V_D$, where V_S are storage zones (or slots), V_J are aisle junctions and V_D are depots/docks; E are traversable aisle segments; $w: E \rightarrow \mathbb{R}_{\geq 0}$ is travel time (length / speed, optionally plus a congestion term). Let $o \in V_D$ be the depot. Let $I = \{1, \dots, n\}$ be candidate inventory items (SKUs or replenishment lots) with:

- $a_i \in \mathbb{N}$ — space requirement (slots or volume units);
- $\lambda_i \geq 0$ — expected pick frequency per planning period;
- r_i — expected service value per period if item i is available in the forward area (for example margin protected or penalty avoided);
- h_i — holding cost per period;
- $C_z \in \mathbb{N}$ — forward-storage capacity of zone $z \in V_S$;
- ρ — cost per unit of travel time; $D_z = d_G(o, z)$ — shortest-path time from the depot to zone z .

Decision variables. $x_{iz} \in \{0,1\}$ equals 1 if item i is stocked in zone z ; $k_z \in \mathbb{N}$ is the number of units of a scarce resource (for example pickers or AGVs) assigned to zone z , with total budget K .

Net value of placing item i in zone z (additive approximation).

$$v_{iz} = r_i - h_i - 2 \cdot \rho \cdot \lambda_i \cdot D_z$$

The factor 2 reflects a round trip from the depot. This model assumes that tour cost is additive across items; real picking tours share travel (batching), so it over-estimates routing cost and is a deliberate simplification (see Section 16).

Joint problem P0 (inventory placement under capacity).

$$\max \sum_i \sum_z v_{iz} \cdot x_{iz} \quad \text{s.t.} \quad \sum_z x_{iz} \leq 1 \quad \forall i; \quad \sum_i a_i \cdot x_{iz} \leq C_z \quad \forall z; \quad x_{iz} \in \{0,1\}$$

P0 is an instance of the generalised assignment problem. With a single zone, D_z is constant and P0 reduces to the 0/1 knapsack problem, so P0 is NP-hard; GAP and multidimensional knapsack problems are noted as strongly NP-hard in [16]. Exact DP is therefore pseudo-polynomial in capacity for one zone, and a decomposition is needed for many zones (Section 12).

Resource-allocation problem P1. Given zone benefit functions $g_z(k)$ (for example expected picks served per period with k pickers in zone z , net of congestion),

$$\max \sum_z g_z(k_z) \quad \text{s.t.} \quad \sum_z k_z \leq K, \quad k_z \in \{0, \dots, K_z\}$$

where g_z need not be concave. P1 is a multiple-choice knapsack: each zone is a group from which exactly one option k is chosen.